higher education & training

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REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

ENGINEERING SCIENCE N4

(15070434)

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09:00–12:00

This question paper consists of 9 pages, 1 formula sheet and 1 information sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
ENGINEERING SCIENCE N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only a BLUE or BLACK pen.
6. Sketches must be large, neat and fully labelled.
7. Use $g = 9{,}8\ m/s^2$
8. Write neatly and legibly.

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SECTION A: GENERAL

QUESTION 1

1.1 Give ONE example of a scenario which explains Newton's first law of motion.

NOTE: Do not state the law. (2)

1.2 Define the following

1.2.1 Angular acceleration

1.2.2 Stress

(2 × 1) (2)

1.3 Show the relationship between:

1.3.1 The linear, area and volume coefficient of expansion (2)

1.3.2 Volume, pressure and temperature in the combination of Boyle's and Charles' gas laws (1)

1.4 State Hooke's law. (2)

[9]

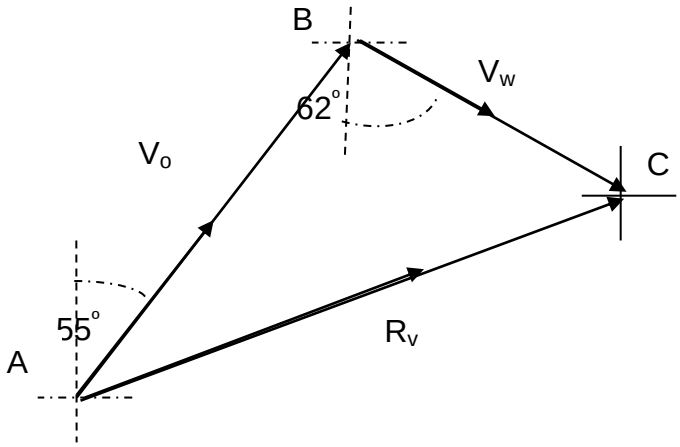
QUESTION 2

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (2.1–2.9) in the ANSWER BOOK.

2.1 The direction of a southwesterly wind is …

A N 45° W
B S 45° E
C N 45° E
D S 45° W (1)

2.2



This sketch represents a plane which flies at 140 km/h N 55° E and is blown off course by a wind of 80 km/h to E 28° S.

NOTE: Vo is the velocity of the plane; V_w is the velocity of the wind; R_v is the resultant velocity.

2.2.1 The magnitude of the resultant velocity of the plane is …

A 87,470 km/h
B 169,499 km/h
C 190,183 km/h
D 124,434 km/h (2)

2.2.2 The direction of the resultant velocity of the plane is …

A E 22,012° N
B N 77,012° E
C E 22,012° N
D E 67,998° N (2)

2.3 When a gardener pushes a lawn mower on a large lawn with three times as much force, thus producing three times as much acceleration, it is an example of the application of …

A Newton's first law of motion.
B Newton's second law of motion.
C Newton's third law of motion.
D the law of conservation of energy. (1)

2.4 … applies when a rocket is launched to the moon and it keeps on moving non-stop unless it hits another object on the way.

A The law of conservation of energy
B Newton's first law of motion
C Newton's second law of motion
D Newton's third law of motion (1)

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2.5 The following is a characteristic of a liquid:

A A liquid takes the shape of its container.
B A liquid depends on pressure.
C Water and hydraulic fluid can be used only once.
D A liquid has no definite volume. (1)

2.6 The kelvin equivalent of 80 °C is …

A 243 K
B 273 K
C 13 K
D 353 K (1)

2.7 Boyle's gas law states that at a constant …

A temperature, the pressure exerted on a gas is directly proportional to the volume.
B pressure, the volume is directly proportional to the temperature.
C temperature, the pressure is inversely proportional to the volume.
D pressure, the volume is inversely proportional to the temperature. (2)

2.8 A stress of 5 kPa would be exerted on the surface area of a square bar with sides 0,5 m by a force of …

A 982 N.
B 1 250 N.
C 393 N.
D 3 927 N. (2)

2.9 A square container with a volume of $0{,}238 \times 10^{-3}$ m^3 at a temperature of 100 K expands and increases its volume by $1{,}05 \times 10^{-4}$ m^3 while the temperature rises by 150 K. The coefficient of volume expansion of the square container is …

A $26{,}471 \times 10^{-4}$/K
B $29{,}412 \times 10^{-4}$/K
C $88\,235 \times 10^{-4}$/K
D $44{,}118 \times 10^{-4}$/K (2)

[15]

TOTAL SECTION A: 24

SECTION B

QUESTION 3: KINEMATICS

3.1 With military precision a marine hit an approaching enemy warship with a missile, 55 km from the shore. The missile was launched from the ground at 520 km/h at 50° to the ground.

Determine the total time (in minutes) for the missile to hit the warship 55 km away (on a horizontal plane). (4)

3.2 A fighter plane Q, flies at 250 km/h east while the velocity of fighter plane P is 210 km/h W 53° S

Determine the velocity of fighter plane ${}_{P}V_{Q}$. (5)
[9]

QUESTION 4: ANGULAR MOTION

4.1 Calculate the rotational frequency (in r/min) of the wheels of a train if it moves at a velocity of 280 km/h.

NOTE: The diameter of a wheel of a train is 720 mm. (2)

4.2 The speed of a flywheel increases from 444 r/min to 2 840 r/min in 0,58 minutes.

Calculate the following:

4.2.1 The angular acceleration of the flywheel

4.2.2 The number of revolutions made during this acceleration

(2 × 3) (6)
[8]

QUESTION 5: DYNAMICS

A vehicle with a mass of 980 kg accelerates uniformly up a slope of 1:45 from 6 m/s to 21 m/s within 35 seconds. The resistance to motion is 0,459 N/kg.

Calculate the following:

5.1 The acceleration of the vehicle (2)

5.2 The accelerating force of the vehicle (2)

5.3 The total force (4)
[8]

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QUESTION 6: STATICS

6.1 A beam of 10 m is supported by A and B which are 10 m apart. A is on the left side of the beam. On the left side of the beam is a 50 N/m distributed load of 5 m. 2 m from the right side of the beam is a concentrated load of 40 N.

NOTE: A = 94,5 N; B = 195,5 N

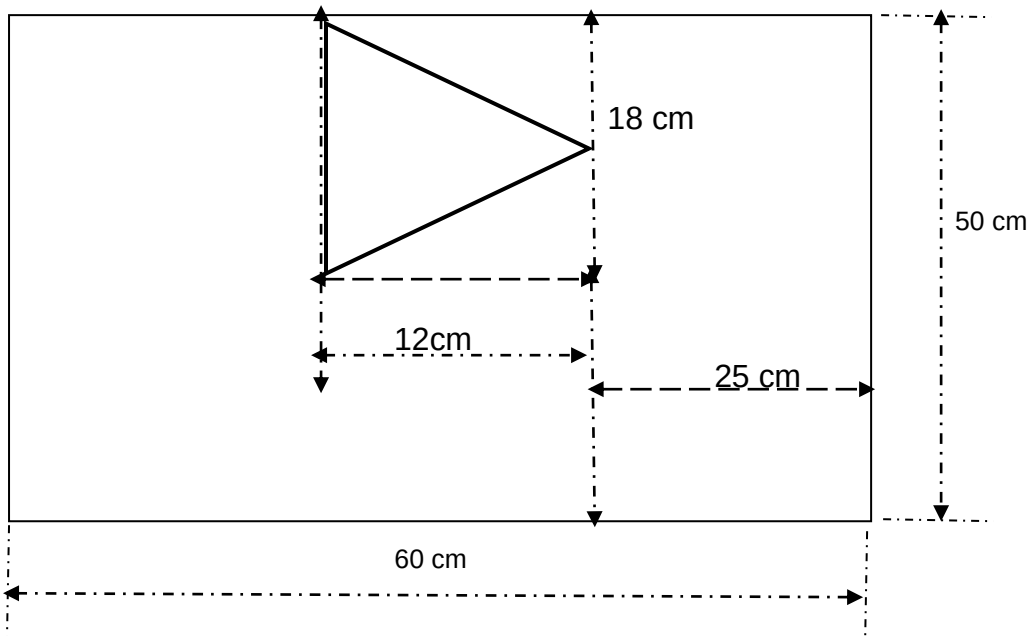
6.1.1 Make a neat, labelled sketch of the beam in detail. (2)

6.1.2 Draw a neat, shear force diagram in detail. (2)

6.1.3 Determine the bending moments at the principal points. (3)

6.1.4 Draw a bending moment diagram in detail. (3)

6.2



This figure shows a lamina with a triangular hole with a base of 12 cm.

Determine the position of the centre of the lamina from the y-axis. (X) (5)
[15]

TOTAL SECTION B: 40

SECTION C

QUESTION 7: HYDRAULICS

7.1 An irrigation system fills an empty circular dam with a diameter of 50 m and a height of 28 m at a pressure of 250 kPa.

Determine the following:

7.1.1 The amount of work needed to fill the dam to capacity (3)

7.1.2 The time (in hours) it will take to drain the dam if a power of 840 kW was applied (2)

7.2 The diameter of a ram is three times that of the plunger, while the stroke length of the plunger is 110 mm. The mechanical advantage of the lever is 22.

Calculate the force that should be applied by the lever to affect the lifting of a load of 2,2 Mg if the efficiency is 91%. (5)

7.3 A double-cylinder single-action piston pump with a diameter of 70 mm and a stroke length of 150 mm, is designed to deliver water at a rate of 0,027 litres per second.

Determine the operating speed (in r/min) if the efficiency is 94%. (4)

[14]

QUESTION 8: STRESS, STRAIN AND YOUNG'S MODULUS OF ELASTICITY

8.1 A round bar with a diameter of 32 mm and a length of 820 mm is subjected to a tensile force of 52,332 kN which increases its length by 0,621 mm.

Calculte the following:

8.1.1 The stress in the bar (3)

8.1.2 The strain (2)

8.1.3 Young's modulus of elasticity of the material of the bar (2)

8.2 A cage with a mass of 1,8 tons is lowered down a shaft by a wire with a diameter of 5 mm and a length of 50 m. The wire stretches by 8 mm.

Determine Young's modulus for the material of the wire. (3)

[10]

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QUESTION 9: HEAT

9.1 A container with dimensions of 500 mm × 200 mm × 600 mm at a temperature of 20 °C is heated and the temperature increases by 245 °C. The linear coefficient of expansion of the material of the container is $12,5 \times 10^{-6}$/K.

Determine the following:

9.1.1 The expansion in length of the longest side (2)

9.1.2 The increase in volume of the container (3)

9.2 A cylinder contains 0,208 m^3 of gas at a pressure of 1 850 kPa and at a temperature of 20 °C. The temperature decreases to 2 °C while the pressure remains constant.

Determine the following:

9.2.1 The volume at 2 °C (2)

9.2.2 The pressure if the volume changes to 0,089 m^3 while the temperature remains at 2 °C (3)

9.2.3 Name the type of gas law applied in QUESTION 9.2.2 and give a reason for your answer. (2)

[12]

TOTAL SECTION C: 36
GRAND TOTAL: 100

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ENGINEERING SCIENCE N4

FORMULA SHEET

Any applicable formula may also be used.

$$L = \frac{u^2}{g}\sin 2\theta$$

$$t_L = 2\frac{u}{g}\sin\theta$$

$$\overline{V} = \frac{s}{t}$$

$$\theta = 2\pi n$$

$$S = R\theta$$

$$\omega = 2\pi N$$

$$\omega = \frac{\theta}{t}$$

$$\omega_2 = \omega_1 + \alpha t$$

$$\omega_2^2 = \omega_1^2 + 2\alpha\theta$$

$$\theta = \omega_1 t + \frac{1}{2}\alpha t^2$$

$$v = \omega R$$

$$v = \pi D n$$

$$a = \alpha R$$

$$\tau = FR$$

$$W_{ork} = \tau\theta = WD$$

$$P = 2\pi n T$$

$$v^2 = u^2 + 2as$$

$$P = T\omega$$

$$n = \frac{N}{60}$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$s = ut + \frac{1}{2}at^2$$

$$P = Fv$$

$$F_a = ma$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$v_{ave} = \frac{u+v}{2}$$

$$P = \frac{F}{A}$$

$$m = \rho \times vol$$

$$P = \rho g h$$

$$\frac{W_r}{F_p} = \frac{D^2}{d^2}$$

$$M.A = \frac{F_p}{F_h}.\frac{100}{\eta} = H.V$$

$$V_s = V_a.\frac{100}{\eta}$$

$$W_{ork}. = P_{ress} \times V_{ol} = A.V.$$

$$Q = mc\Delta t$$

$$\Delta l = l_o\alpha\Delta t$$

$$\beta = 2\alpha$$

$$\gamma = 3\alpha$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$PV = mRT$$

$$\in = \frac{x}{l}$$

$$E = \frac{\sigma}{\in}$$

$$\sigma = \frac{F}{A}$$

$$E = \frac{Fl}{Ax}$$

$$\bar{y} = \frac{A_1y_1 \pm A_2y_2 \ldots}{A_1 \pm A_2 \ldots}$$

$$\bar{y} = \frac{v_1y_1 \pm v_2y_2 \ldots}{v_1 \pm v_2 \ldots}$$

Centroid of half circle 0,424 r

Centroid of triangle is $C = \frac{h}{3}$

Centre of gravity of half circle is $G = \frac{3}{8}r$

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INFORMATION SHEET

PHYSICAL CONSTANTS

QUANTITY	CONSTANTS
Atmospheric pressure	101,3 kPa
Density of copper	8 900 kg/m^3
Density of aluminium	2 770 kg/m^3
Density of gold	19 000 kg/m^3
Density of alcohol (ethyl)	790 kg/m^3
Density of mercury	13 600 kg/m^3
Density of platinum	21 500 kg/m^3
Density of water	1 000 kg/m^3
Density of mineral oil	920 kg/m^3
Density of air	1,05 kg/m^3
Electrochemical equivalent of silver	1,118 mg/C
Electrochemical equivalent of copper	0,329 mg/C
Gravitational acceleration	9,8 m/s^2
Heat value of coal	30 MJ/kg
Heat value of anthracite	35 MJ/kg
Heat value of petrol	45 MJ/kg
Heat value of hydrogen	140 MJ/kg
Linear coefficient of expansion of copper	17 × 10^{-6}/°C
Linear coefficient of expansion of aluminium	23 × 10^{-6}/°C
Linear coefficient of expansion of steel	12 × 10^{-6}/°C
Linear coefficient of expansion of lead	54 × 10^{-6}/°C
Specific heat capacity of steam	2 100 J/kg.°C
Specific heat capacity of water	4 187 J/kg.°C
Specific heat capacity of aluminium	900 J/kg.°C
Specific heat capacity of oil	2 000 J/kg.°C
Specific heat capacity of steel	500 J/kg.°C
Specific heat capacity of copper	390 J/kg.°C

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